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Semiconductor chronotomography: A lifetime demultiplexing and imaging method for revealing hidden carrier loss mechanisms in electronic solids (invited)

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ABSTRACT

The concept of semiconductor chronotomography (CTG) is introduced as a scanning adaptive frequency gate, a process that significantly enhances radiative [photocarrier radiometry (PCR) and its camera imaging extension, lock-in carrierography (LIC)] or non-radiative (photothermal radiometry and its camera imaging extension, lock-in thermography) signal responses to the presence of multiple free-carrier lifetime limiting mechanisms. When in a laser photoexcitation modulation frequency scan, the free-carrier decay rate matches/lies within the gatewidth between two adaptive successive modulation frequencies, CTG yields signal extrema related to resonances with free-carrier decay or recombination rates. CTG was successfully applied to the investigation of multiple lifetimes in *p*- and *n*-type Si wafers and mono- and multicrystalline solar cells using homodyne and heterodyne PCR and LIC modalities. The statistical distribution of relaxation times τ and the product $f(\tau) \times \tau$, where $f(\tau)$ is the probability of occurrence of lifetime τ, are important CTG outputs: the former reveals the impact of each lifetime, while the latter weighs the lifetime with the relative probability of its occurrence. In summary, CTG in conjunction with PCR is a highly sensitive resonant rate modality. Its high sensitivity reveals multiple lifetime distributions, crucially short lifetimes otherwise hidden below much longer carrier recombination lifetimes.

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I. INTRODUCTION

Photocarrier radiometry (PCR)[1] and its imaging counterpart, lock-in carrierography (LIC),[2,3] are non-contact and non-destructive spectrally gated frequency-domain dynamic photoluminescence (PL) techniques. They have been applied extensively to non-contact simultaneous determination of electronic transport parameters such as trap states/activation energies,[4] lifetimes, ambipolar diffusivities, front and back surface recombination velocities, interface recombination velocities,[5,6] minority carrier lifetimes,[6,7] local distributions of resistivity[8] in semiconductor substrates and solar cells through quantitative amplitude- and phase frequency response measurements, and theoretical carrier diffusion wave (CDW) models to extract transport parameters.[9,10] Conventional PCR in which the exciting optical beam is modulated at a single frequency is referred to as "homodyne" (HoPCR). More recently, two-beam heterodyne PCR (HePCR) has been introduced to address the need for high-frequency responses required to

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measure recombination lifetimes and other fast photocarrier relaxation processes.[11] Similar developments such as relaxation times and surface recombination velocities have been reported with regard to the introduction of heterodyne LIC (HeLIC) imaging[11,12] to overcome the NIR camera low frame-rate (image capture speed) limitations. A newly observed heterodyne HePCR frequency response phenomenon in the form of amplitude depression (also termed a "dip" or "notch"), accompanied by a 180° phase transition, has been discovered and was shown to be a nonlinear effect attributed to trap-state-related interactions between the harmonically excited CDW and defect states or traps in Si wafers.[13] This phenomenon has been attributed to several free photocarrier decay processes associated with defect and/or trap states[13,14] and has raised issues of fast rate-limiting lifetimes when simultaneously present along with dominant radiative and nonradiative recombination processes. In this work, the concept of chronotomography (CTG; literally "time-slicing" from the Greek "chronos," time, and "tomographia," slice imaging) is introduced as an important analysis tool for semiconductor optoelectronic diagnostics and quality control. CTG reveals efficient carrier loss mechanisms with short lifetimes, which are otherwise hidden below much longer, often less probable, yet more readily measurable lifetimes. In the first instance, CTG is applied to the measurement of "hidden" relaxation times in n- and p-type Si wafers and a Si solar cell using frequency scanned PCR and NIR camera homodyne LIC (HoLIC) and HeLIC imaging.

II. SEMICONDUCTOR LIFETIME CHRONOTOMOGRAPHY

A. Chronotomography theory

Semiconductor materials often exhibit a series of free-carrier decay rates due to defects, traps, and various recombination pathways, leading to a distribution of carrier lifetimes. A time-domain approach to photoluminescence (radiative emission, PL) collective decay kinetics of free photoexcited carriers undergoing multiple types of de-excitation in the presence of a continuum of trap states, including recombining across the bandgap, can be described as[15]

$$I(t) = \int_{\tau_{\min}}^{\tau_{\max}} f(\tau) e^{-t/\tau} d\tau, \tag{1}$$

where $f(\tau)$ is the lifetime distribution function with physical meaning that of a probability function for a particular lifetime τ to occur. $[\tau_{\max}, \tau_{\min}]$ represents the full range of decay time constants involved in a given carrier de-excitation process. The frequency dependence of modulated photoluminescence, the radiative-decay-associated (as opposed to, e.g., non-radiative decay) spectrally gated portion of which constitutes frequency-domain photocarrier radiometry (FD-PCR or simply PCR),[1] can be described from the Fourier transform $F(\omega)$ of $I(t)$ in Eq. (1) obtained by extending the limits of $[\tau_{\max}, \tau_{\min}] \rightarrow (\infty, 0]$,

$$F(\omega) = \int_0^{\infty} \frac{\tau f(\tau)}{1 + i\omega\tau} d\tau. \tag{2}$$

When discrete lifetime distributions are involved, the distribution function can be written as

$$f(\tau) \rightarrow \sum_{j=1}^{N} f(\tau)\delta(\tau - \tau_j), \tag{3}$$

and Eq. (2) simplifies to the sum of discrete contributions from all decay lifetimes,

$$F(\omega) = \sum_{j=1}^{N} f(\tau_j)\left(\frac{\tau_j}{1 + i\omega\tau_j}\right), \tag{4}$$

where N is the number of the effective multi-lifetime components and $(f(\tau_j))$ takes on the meaning of occurrence probability of the jth lifetime τ_j.

The concept of frequency-domain chronotomography (FD-CTG or simply CTG) is that of a scanning adaptive frequency gate, a resonant process that significantly enhances the PCR signal response to the presence of a free-carrier lifetime limiting mechanism when the free-carrier decay rate matches/lies within the frequency gatewidth between two successive modulation frequencies (a frequency gate). The angular frequency gatewidth $\Delta\omega$ is defined between two adjacent values ω_1 and ω_2 as

$$\begin{gathered}\omega_1 + \Delta\omega = \omega_2 \equiv k\omega_1 \Rightarrow \Delta\omega = (k-1)\omega_1; \\ k = \omega_2/\omega_1 > 1.\end{gathered} \tag{5}$$

The enhanced sensitivity of this process to carrier lifetimes stems from the fact that the scanned frequency gate acts like a derivative operator in that the PCR signal exhibits an extremum if the statistical decay rate of a carrier de-excitation event such as, e.g., radiative emission from, or capture into, an electronic defect state or trap, lies within the scanning gatewidth. Specifically, the free-carrier decay signature within $\Delta f = \Delta\omega/2\pi$ acts like a resonant frequency peak or trough, depending on the sign of the expression inside the parentheses of the following equation:

$$\begin{aligned}\Delta F(\omega) &= \Delta F(\omega_1, \omega_2, \ldots, \omega_j, \ldots, \omega_N) \\ &= \sum_{j=1}^{N} \tau_j f(\tau_j)\left(\frac{1}{1 + ik\omega_j\tau_j} - \frac{1}{1 + i\omega_j\tau_j}\right) \equiv \sum_{j=1}^{N} \Delta F(\omega_j).\end{aligned} \tag{6}$$

It is evident that the narrower the Δf, the sharper and more highly localized the extremum will be when the condition $\tau \sim 1/\Delta f$ is satisfied: When such a condition occurs within a narrow Δf, the peak must by necessity be sharp enough to fit within Δf. The extrema of the function $\Delta F(\omega)$, the signal difference between frequencies ω_j and $\omega_j + \Delta\omega_j$, can be obtained variationally by extremizing $\Delta F(\omega)$. Upon treating $\Delta F(\omega)$ as a stationary functional in variational calculus and employing the

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Rayleigh–Ritz method,[16] the expression

$$d\Delta F(\omega) = d\Delta F(\omega_1, \omega_2, \ldots, \omega_j, \ldots, \omega_N) = \left(\frac{\partial \Delta F}{\partial \omega_1}\right)\Delta\omega_1 + \left(\frac{\partial \Delta F}{\partial \omega_2}\right)\Delta\omega_2 + \ldots + \left(\frac{\partial \Delta F}{\partial \omega_j}\right)\Delta\omega_j + \ldots + \left(\frac{\partial \Delta F}{\partial \omega_N}\right)\Delta\omega_N = 0 \tag{7}$$

denotes the stationary values (i.e., the extrema) of $\Delta F(\omega)$. Each term $\left(\frac{\partial \Delta F}{\partial \omega_j}\right)$ is then set equal to 0 because all $\{\omega_j\}$ values at which extrema occur are independent of each other under variation $\Delta\omega_j$,

$$\frac{\partial \Delta F}{\partial \omega_1} = \frac{\partial \Delta F}{\partial \omega_2} = \ldots = \frac{\partial \Delta F}{\partial \omega_j} = \ldots = \frac{\partial \Delta F}{\partial \omega_N} = 0. \tag{8}$$

As the frequency sweep is performed over a range $[\omega_{\min}, \omega_{\max}]$, signal extrema can be encountered such that for each extremum, i.e., for each τ_j, Eqs. (6) and (8) yield

$$\frac{d}{d\omega}\Delta F(\omega) = \frac{d}{d\omega}\sum_{j=1}^{N}\tau_j f(\tau_j)\left(\frac{1}{1+ik\omega\tau_j} - \frac{1}{1+i\omega\tau_j}\right) = 0 \Rightarrow \frac{d}{d\omega}\left(\frac{1}{1+ik\omega\tau_j} - \frac{1}{1+i\omega\tau_j}\right) = 0. \tag{9}$$

For radiative de-excitations of free photocarriers, given that the PCR signal is complex, the condition for extremization can be applied to both amplitude and phase frequency scanning gates and turns out to be different for each of these two signal channels.

1. Amplitude extrema

Writing the complex expressions in Eq. (9) in polar coordinate notation

$$\frac{1}{1+i(k)\omega\tau_j} \equiv Z_j = |Z_j|e^{i\Phi_j} \tag{10a}$$

with

$$|Z_j[(k)\omega]| = \frac{1}{\sqrt{1+[(k)\omega\tau_j]^2}};\ \Phi_j = \tan^{-1}[-(k)\omega\tau_j], \tag{10b}$$

the terms inside the r.h.s. parentheses in Eq. (9) can be written as

$$\frac{d}{d\omega}[|Z_j(k\omega)| - |Z_j(\omega)|] = 0, \tag{11}$$

which yields

$$-\frac{k^2\tau_j^2\omega}{\left[1+(k\omega\tau_j)^2\right]^{\frac{3}{2}}} + \frac{\tau_j^2\omega}{\left[1+(\omega\tau_j)^2\right]^{\frac{3}{2}}} = 0. \tag{12}$$

Therefore, a lifetime–angular frequency relation can be derived from the amplitude extrema as follows:

$$\tau_j = \left(\frac{k^{2/3}+1}{k^{4/3}}\right)^{1/2}\frac{1}{\omega^*_{(A),j}}, \tag{13}$$

where $\omega^*_{(A),j}$ denotes the angular frequency of the jth extremum. The amplitude $|\Delta F_j(\omega^*_{(A),j})|$ at the extremum is then found to be proportional to the product of lifetime and probability of its occurrence, $f(\tau_j)\tau_j$,

$$|\Delta F_j\left(\omega^*_{(A),j}\right)| = f(\tau_j)\tau_j\left|\frac{1}{\sqrt{1+\left(k\tau_j\omega^*_{(A),j}\right)^2}} - \frac{1}{\sqrt{1+\left(\tau_j\omega^*_{(A),j}\right)^2}}\right| \equiv |C(k)|f(\tau_j)\tau_j. \tag{14}$$

Here, $C(k)$ is a constant pre-factor common to all extrema and solely determined by the frequency gate width k,

$$C(k) = \frac{1-k^{2/3}}{\sqrt{k^{4/3}+k^{2/3}+1}}. \tag{15}$$

Finally, collecting all lifetime amplitude terms across the entire angular modulation frequency range yields

$$|\Delta F\left(\omega^*_{(A)}\right)| = \sum_{j=1}^{N}|\Delta F\left(\omega^*_{(A),j}\right)| = |C(k)|\sum_{j=1}^{N}\tau_j f(\tau_j). \tag{16}$$

In terms of PCR signals, Eq. (13) implies that in plots of $|\Delta F(\omega)|$ vs $\left(\frac{k^{4/3}}{k^{2/3}+1}\right)^{1/2}\omega$, a relaxation time axis can be constructed such that

$$\tau(\omega) = \left(\frac{k^{2/3}+1}{k^{4/3}}\right)^{1/2}\frac{1}{\omega}, \tag{17}$$

with amplitude extrema emerging at $\omega = \omega^*_{(A),j}; j = 1, \ldots, N$.

These plots can be expressed equivalently vs $\tau(\omega) \Leftrightarrow \omega^{-1}$. When used in a lock-in carrierography (LIC) imaging modality,[17] a LIC $|\Delta F(\omega)|$ image tomogram may exhibit extrema $\tau(\omega^*_{(A),j})$ during an ω scan at particular coordinate locations, L, corresponding to the condition $\omega = \omega^*_{(A),L} \Leftrightarrow \tau_L^{-1}$ being met. For the selected value of k, the amplitude at these extrema locations will be equal to $|C(k)|f(\tau_L)\tau_L$. This process of quantitatively imaging local multi-lifetimes defines the FD-CTG method.

2. Phase extrema

In a manner similar to amplitude extremization, the phase extrema can be found from the condition,

$$\frac{d}{d\omega}[\Phi_j(k\omega) - \Phi_j(\omega)] = 0, \tag{18}$$

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for all (j), or equivalently,

$$\frac{d}{d\omega}\left[-\tan^{-1}(k\omega\tau_j)+\tan^{-1}(\omega\tau_j)\right]=0\Rightarrow\frac{k\tau_j}{1+(k\omega\tau_j)^2}-\frac{\tau_j}{1+(\omega\tau_j)^2}=0. \tag{19}$$

This equation shows that a somewhat different lifetime–angular frequency relation from that of amplitude can be derived from the phase extrema as follows:

$$\tau_j=\left(\frac{1}{k^{1/2}}\right)\frac{1}{\omega^*_{(\phi),j}}, \tag{20}$$

with phase extrema emerging at $\omega=\omega^*_{(\phi),j}$; $j=1,\ldots,N$. From the phase frequency gate,

$$\Delta\Phi\left(\omega_j^*\right)=\Phi_j\left(k\omega_j^*\right)-\Phi_j\left(\omega_j^*\right)=-\tan^{-1}\left(k\omega_j^*\tau_j\right)+\tan^{-1}\left(\omega_j^*\tau_j\right)$$
$$=-\tan^{-1}\left(k^{\frac{1}{2}}\right)+\tan^{-1}\left(\frac{1}{k^{\frac{1}{2}}}\right). \tag{21}$$

Using the trigonometric identity $\tan^{-1}(x)=\frac{\pi}{2}-\tan^{-1}\left(\frac{1}{x}\right)$, the CTG phase at $\omega=\omega^*_{(\phi),j}$ can be obtained,

$$\Delta\Phi\left(\omega_j^*\right)=\frac{\pi}{2}-2\tan^{-1}\left(k^{\frac{1}{2}}\right), \tag{22}$$

which is an expression independent of τ_j and/or $f(\tau_j)$.

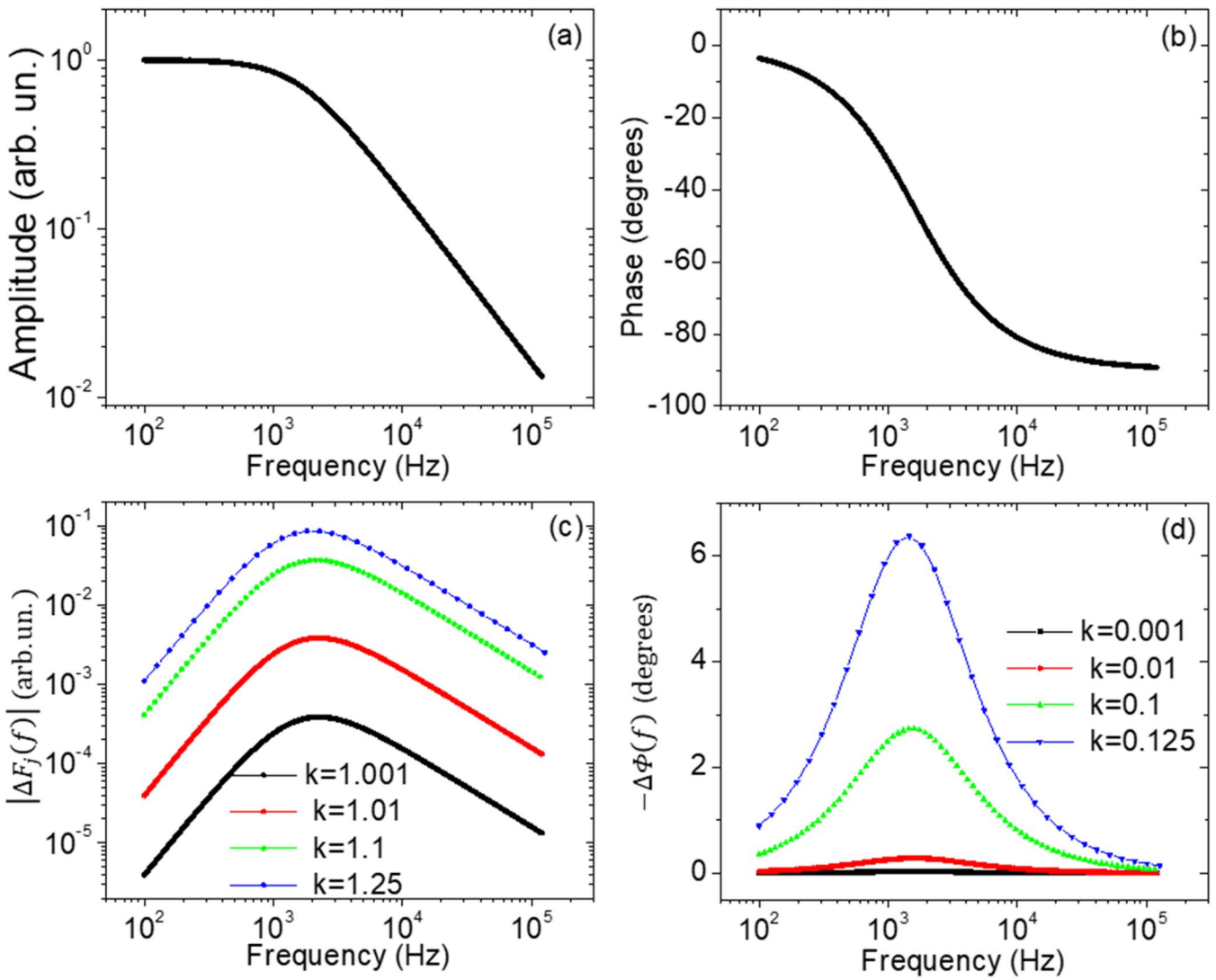


FIG. 1. (a) Simulated amplitude, (b) phase, (c) $|\Delta F_j(f)|$, and (d) $-\Delta\Phi(f)$ frequency dependencies. Parameters: $\tau=100\,\mu$s, $f(\tau)=10^4$. $|\Delta F_j(\omega)|$ and $\Delta\varnothing(\omega)$ are calculated with $k=1.001$, 1.01, 1.1, 1.25.

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TABLE I. Estimated frequency extrema and corresponding single lifetimes extracted from amplitude (A) and phase (Φ) for various k values.

	$k = 1.001$	$k = 1.01$	$k = 1.1$	$k = 1.25$
$f_{ex}^{(A)}$ (Hz)	2250	2240	2155	2073
$f_{ex}^{(\Phi)}$ (Hz)	1591	1584	1523	1431
$\tau^{(A)}$ (s) of $f_{ex}^{(A)}$	1×10^{-4}	1×10^{-4}	9.96×10^{-5}	9.72×10^{-5}
$\tau^{(\Phi)}$ (s) of $f_{ex}^{(\Phi)}$	1×10^{-4}	1×10^{-4}	9.96×10^{-5}	9.95×10^{-5}
$f(\tau^{(A)})$	1×10^{4}	1×10^{4}	1×10^{4}	1.02×10^{4}

Similarly to the CTG amplitude, phase plots vs $\tau(\omega) \Leftrightarrow \omega^{-1}$ can be constructed and extrema will appear at angular frequencies satisfying Eq. (20).

B. Simulations

Within the scanning adaptive frequency gate CTG amplitude and phase channels, the resolution of $|\Delta F(\omega)|$, $\Delta\Phi(\omega_j^*)$ and their extrema is related to the value of the k parameter, which controls the $\Delta\omega$ gatewidth. The amplitude and phase dependencies were

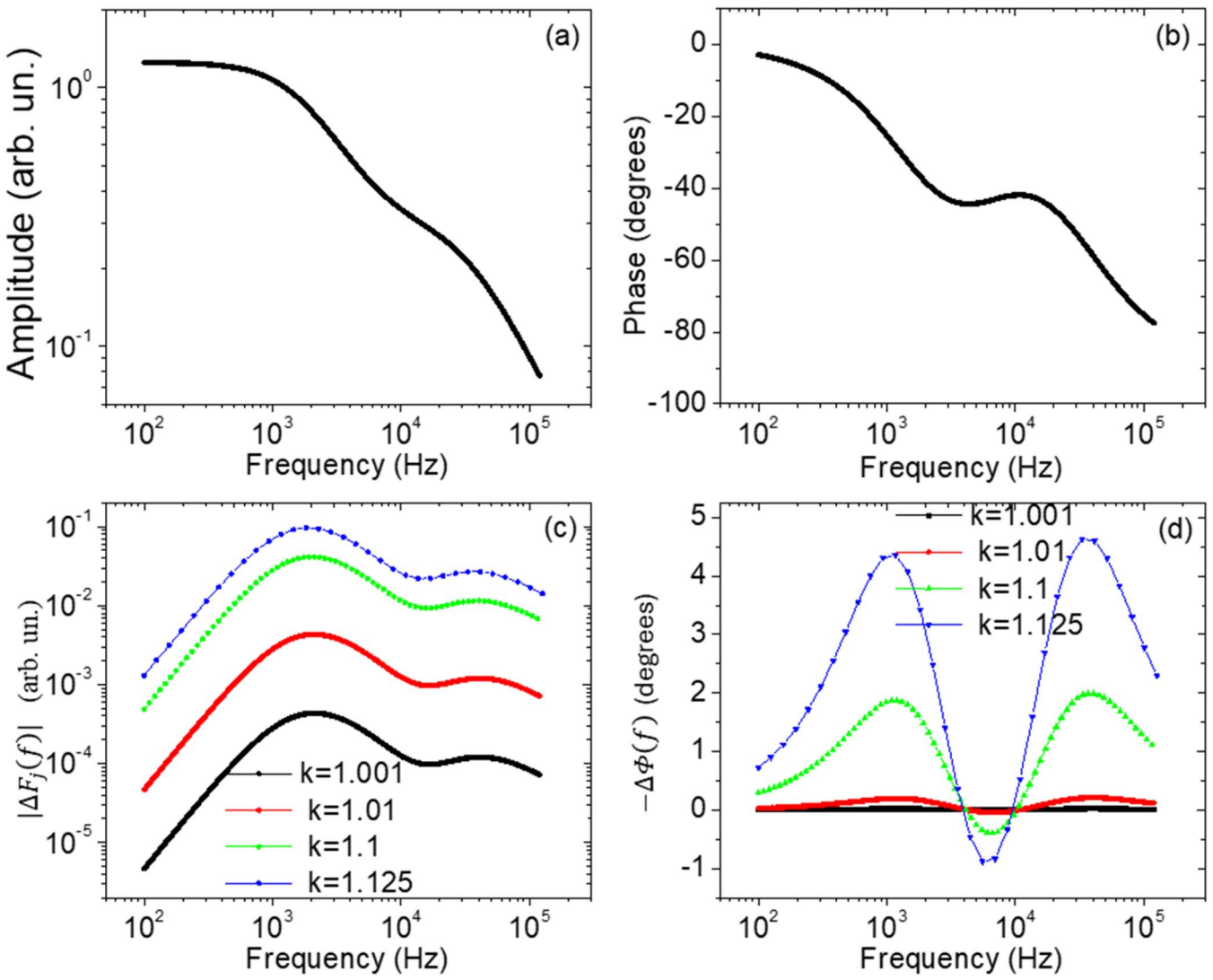


FIG. 2. (a) Simulated amplitude, (b) phase, (c) $|\Delta F_j(f)|$, and (d) $-\Delta\Phi(f)$ frequency dependencies. Parameters: $\tau_1 = 100\,\mu$s, $f(\tau_1) = 10^4$, $\tau_2 = 5\,\mu$s, $f(\tau_2) = 5 \times 10^4$. $|\Delta F_j(\omega)|$ and $\Delta\Phi(\omega)$ were calculated with $k = 1.001, 1.01, 1.1, 1.25$.

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TABLE II. Estimated frequency extrema and corresponding two lifetimes for various k values.

	$k = 1.001$	$k = 1.01$	$k = 1.1$	$k = 1.25$
$f_{ex\,1}^{(A)}$ (Hz)	2113	2104	2025	1927
$f_{ex\,2}^{(A)}$ (Hz)	41 145	40 963	39 334	37 307
$f_{ex\,1}^{(\varnothing)}$ (Hz)	1213	1208	1157	1083
$f_{ex\,1}^{(\varnothing)}$ (Hz)	40 264	40 087	38 662	37 348
$\tau_1^{(a)}$ (s) of $f_{ex\,1}^{(A)}$	1.06×10^{-4}	1.06×10^{-4}	1.06×10^{-4}	1.05×10^{-4}
$\tau_2^{(a)}$ (s) of $f_{ex\,2}^{(A)}$	5.47×10^{-6}	5.47×10^{-6}	5.46×10^{-6}	5.4×10^{-6}
$\tau_1^{(\varnothing)}$ (s) of $f_{ex\,1}^{(\Phi)}$	1.31×10^{-4}	1.31×10^{-4}	1.31×10^{-4}	1.31×10^{-4}
$\tau_2^{(\varnothing)}$ (s) of $f_{ex\,2}^{(\Phi)}$	3.95×10^{-6}	3.95×10^{-6}	3.93×10^{-6}	3.81×10^{-6}
$f(\tau_1^{(a)})$	1.06×10^{4}	1.06×10^{4}	1.06×10^{4}	1.07×10^{4}
$f(\tau_2^{(a)})$	5.73×10^{4}	5.73×10^{4}	5.73×10^{4}	5.77×10^{4}

simulated with parameters $\tau = 100\,\mu s$ and $f(\tau) = 10^4$. This value of $f(\tau)$ is arbitrary and can be normalized to a maximum of unity, consistent with its interpretation as a probability of occurrence of a given lifetime. The frequency gatewidth was constructed as in Eq. (5): $f_{i+1} = kf_i$; $f_i = \omega_i / 2\pi$. The simulated PCR amplitude and phase results from Eq. (4), plotted vs modulation frequency, are shown in Figs. 1(a) and 1(b), respectively. The corresponding $|\Delta F_j(\omega)|$ and $\Delta\Phi(\omega)$ are shown in Figs. 1(c) and 1(d). Both $|\Delta F_j(\omega)|$ and $\Delta\Phi(\omega)$ demonstrate very clear peaks related to lifetime.

The precise extrema positions were estimated according to $\frac{d}{d\omega}\Delta F(\omega) = 0$ using a linear frequency increment approach between adjacent points. The estimated extremum frequencies and corresponding lifetimes for various values of k are presented in Table I. The lifetimes estimated from the CTG maxima coincide within two significant decimal figures with the assumed simulated value for all k values near unity, in both amplitude and phase plots, with slight discrepancies for higher k values.

A simulation of two lifetimes with $\tau_1 = 100\,\mu s$, $f(\tau_1) = 10^4$, $\tau_2 = 5\,\mu s$, $f(\tau_2) = 5 \times 10^4$ is shown in Fig. 2 with the estimated frequency extrema and lifetimes in Table II. $|\Delta F_j(\omega)|$ and $\Delta\Phi(\omega)$ exhibit two peaks related to the two assumed lifetimes, with much improved resolution in the phase plot. The lifetimes obtained from the CTG frequency scans were found to be close to the assumed simulation values. The observed differences were caused by overlap interference (a measure of frequency resolution) between the two lifetimes and were also affected by the values of $f(\tau_i)$ as the latter can shift the extrema positions on the frequency axis. Larger discrepancies between the assumed and estimated lifetime values extracted

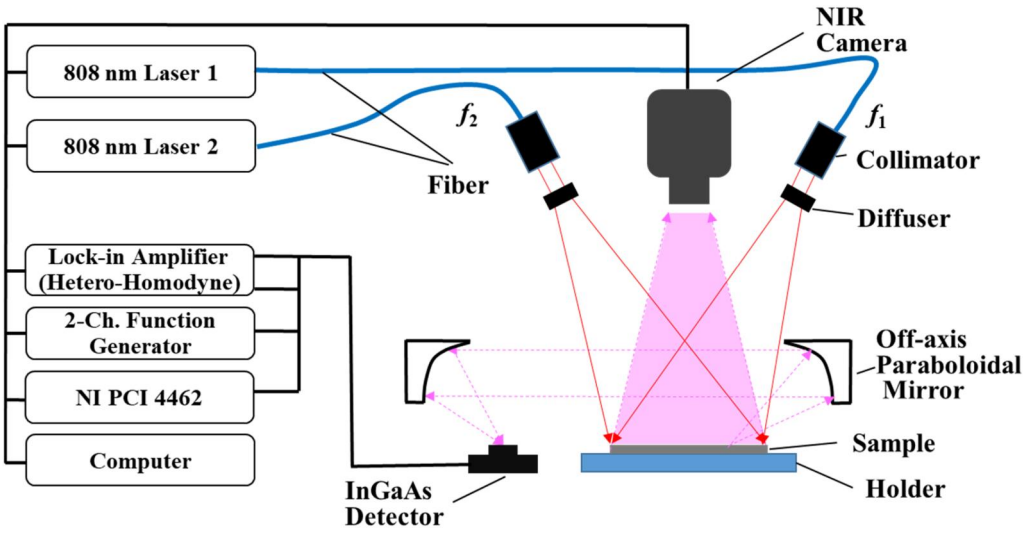


FIG. 3. Schematic combined homodyne/heterodyne PCR/LIC imaging experimental setup.

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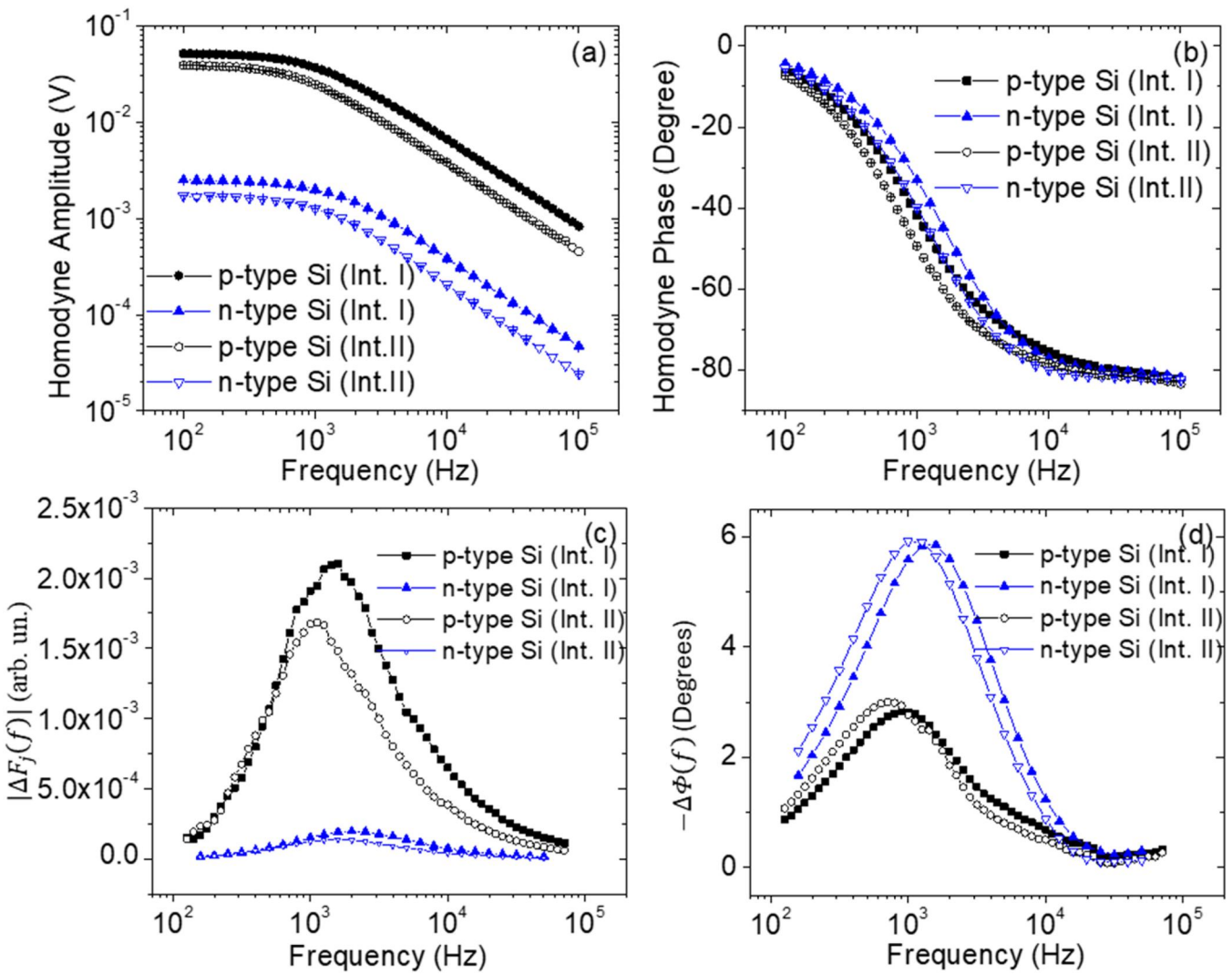


FIG. 4. (a) The homodyne amplitude, (b) phase, (c) $|\Delta F_j(\omega)|$, and (d) $-\Delta\Phi(\omega)$ frequency plots for *p*-type and *n*-type Si wafers at excitation laser beam intensities I and II.

for the phase plots are due to the higher resolution of the CTG phase than the amplitude scan.

Figure 2(c) shows the effects of $f(\tau_i)\tau_i$ dependence of the amplitude $\Delta F(f)$ plots, Eqs. (14)–(16): $\tau_1 f(\tau_1) = 1 \times 10^6\,\mu$s, whereas $\tau_2\, f(\tau_2) = 25 \times 10^4\,\mu$s. The CTG amplitude scales with these values, while the $\Delta\Phi(f)$ plots [Fig. 2(d)] do not depend on them but only on the k value. It is clear from Fig. 2(d) that the phase peaks appear approximately at the same values of $\Delta\Phi$ for a given k.

TABLE III. Estimated extrema frequencies and corresponding lifetimes with weighting factors for the p-type and n-type Si wafers using homodyne PCR at excitation laser intensities I and II.

	k	$f_{ex}^{(A)}$ (Hz)	$f_{ex}^{(\Phi)}$ (Hz)	$\tau^{(A)}$ (s) of $f_{ex}^{(A)}$	$\tau^{(\Phi)}$ (s) of $f_{ex}^{(\Phi)}$
p-type Si (Int. I)	1.122	1476	924	1.44×10^{-4}	1.63×10^{-4}
p-type Si (Int. II)	1.122	1109	728	1.92×10^{-4}	2.06×10^{-4}
n-type Si (Int. I)	1.259	2057	1448	9.77×10^{-5}	9.8×10^{-5}
n-type Si (Int. II)	1.259	1584	1127	1.27×10^{-4}	1.26×10^{-4}

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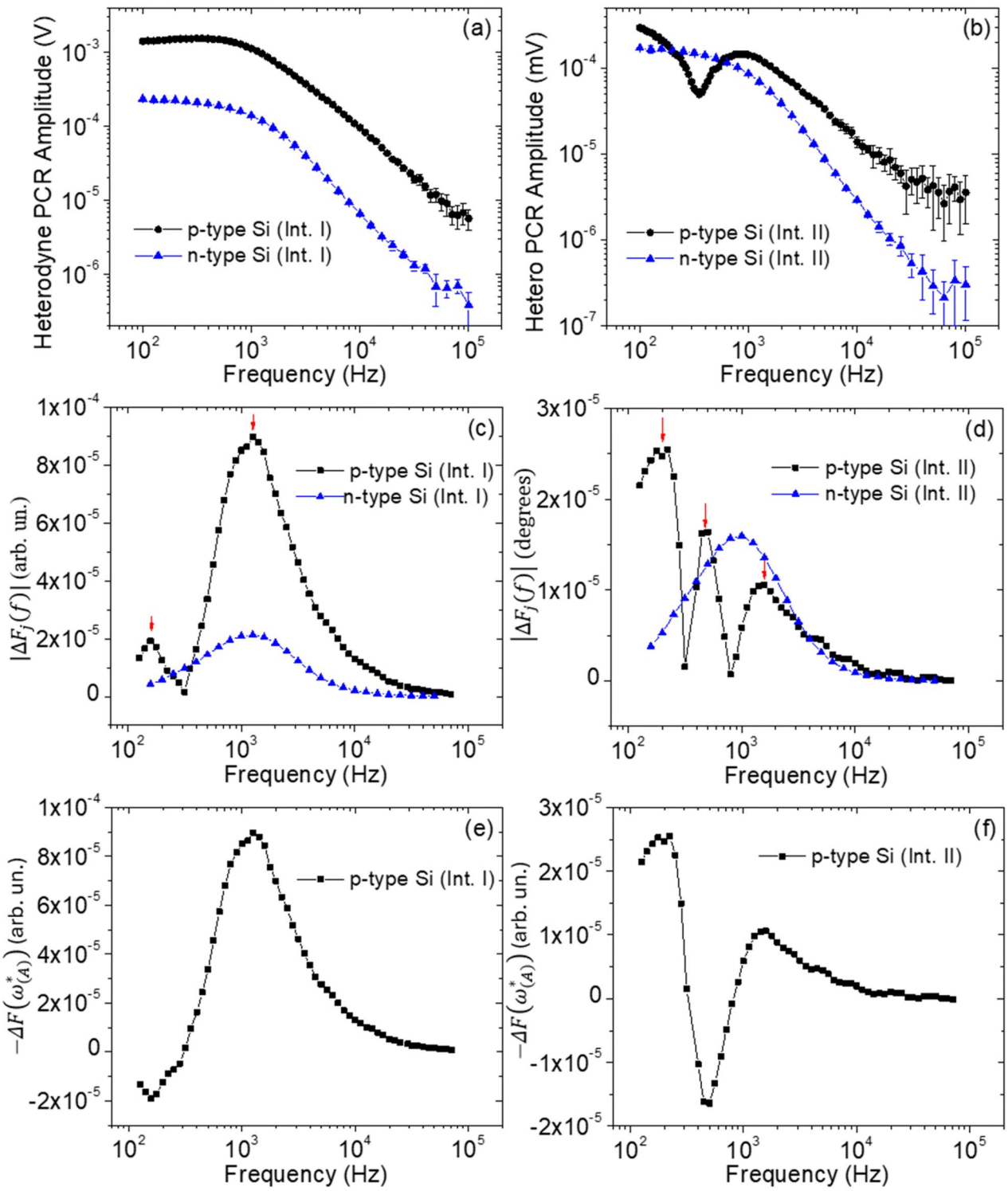


FIG. 5. Heterodyne PCR (HePCR) amplitude of the *p*-type and *n*-type Si wafers (a) and (b), the corresponding $|\Delta F_j(\omega)|$ (c) and (d), and corresponding $-\Delta F(\omega^*_{(A)})$ (e) and (f) for excitation laser intensities I (a) and (c) and II (b) and (d). In (c), the two extrema, and in (d), the three extrema are highlighted using arrows.

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TABLE IV. Estimated extrema frequencies and extracted lifetimes with corresponding weighting factors for *p*-type and *n*-type Si wafers using HePCR.

	k	$f_{ex}^{(A)}$ (Hz) Intensity I	$\tau^{(A)}$ (s) Intensity I	$f(\tau^{(A)})$ Intensity I	$f_{ex}^{(A)}$ (Hz) Intensity II	$\tau^{(A)}$ (s) Intensity II	$f(\tau^{(A)})$ Intensity II
p-type	1.220	160	1.28×10^{-3}	−0.326	…	…	…
	1.220	1296	1.57×10^{-4}	12.3	…	…	…
	1.125	…	…	…	201	1.06×10^{-3}	0.517
	1.125	…	…	…	484	4.39×10^{-4}	−0.818
	1.125	…	…	…	1517	1.4×10^{-4}	1.65
n-type	1.259	1219	1.65×10^{-4}	…	954	2.11×10^{-4}	…

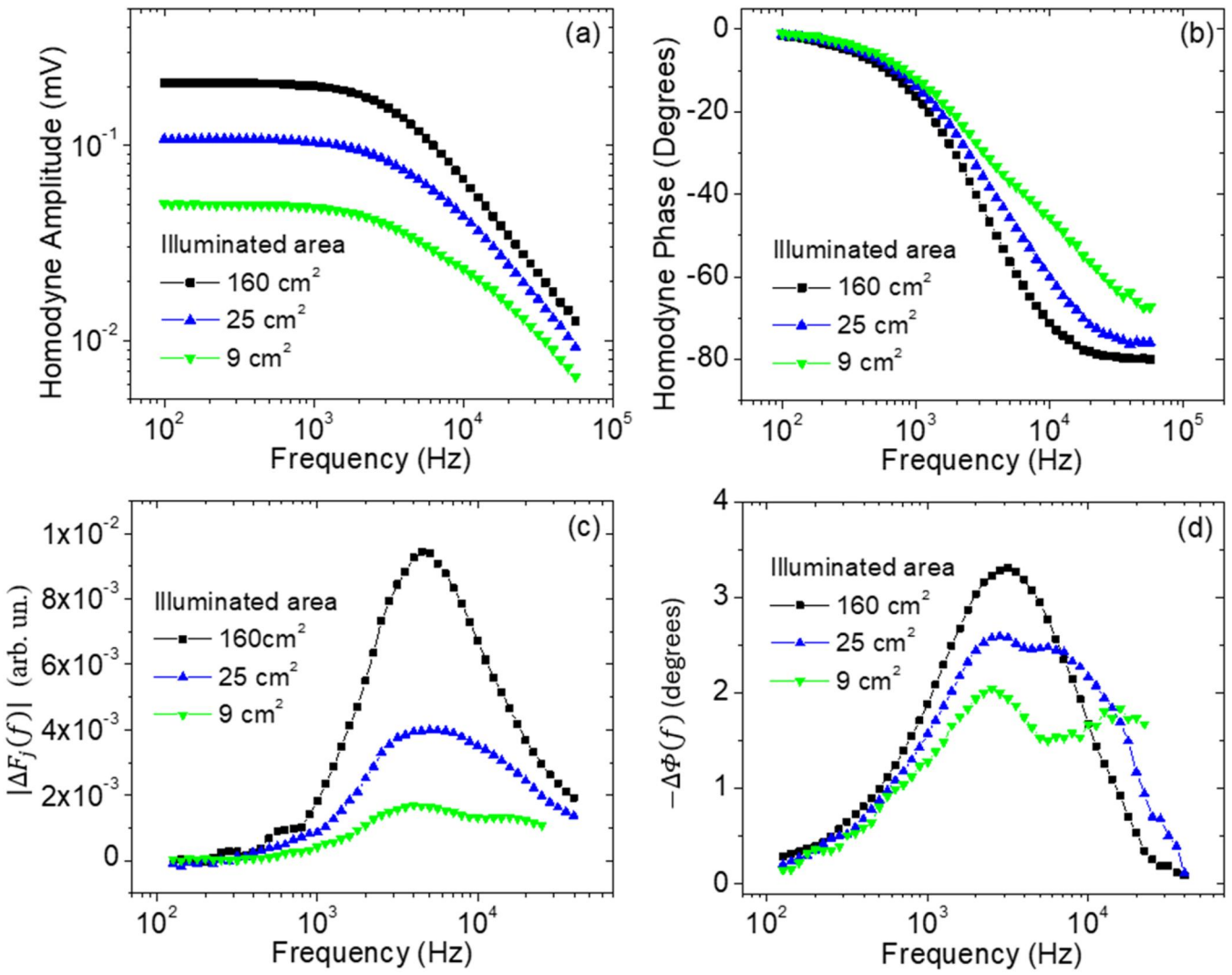


FIG. 6. (a) The HoPCR amplitude, (b) phase, (c) CTG amplitude $|\Delta F_j(\omega)|$, and (d) phase $-\Delta\Phi(\omega)$ frequency plots of *n*-type monocrystalline solar cell under illumination of different areas.

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TABLE V. Estimated extrema frequencies and lifetimes from extrema frequencies with corresponding weighting factors for the *n*-type monocrystalline Si solar cell at various illuminated areas.

Illuminated area (cm^2)	k	$f_{ex}^{(A)}$ (Hz)	$f_{ex}^{(\Phi)}$ (Hz)	$\tau^{(A)}$ (s)	$f(\tau^{(A)})$	$\tau^{(\Phi)}$ (s)
160	1.122	4.61×10^3	3.13×10^3	4.61×10^{-5}	…	4.8×10^{-5}
25	1.122	5.12×10^3	2.84×10^3	4.15×10^{-5}	…	5.29×10^{-5}
	1.122		5.35×10^3			2.81×10^{-5}
9	1.122	4.17×10^3	2.51×10^3	5.10×10^{-5}	7.56×10^2	5.99×10^{-5}
	1.122	1.5×10^4	1.57×10^4	1.42×10^{-5}	2.09×10^3	9.57×10^{-6}

III. MATERIALS AND EXPERIMENTAL

The schematic of the combined experimental PCR/HeLIC system is shown in Fig. 3.

In this system, the beams of two 808 nm fiber-coupled high power lasers with the same parameters were first collimated, then spread and homogenized by microlens arrays to form an illuminated square area, consistent with the size of the Si wafers and solar cell inspected in this work.

The two beams used for spectrally gated PCR excitation were sine- or square-wave modulated using a two-channel function generator in the range 0.1–100 kHz. The modulated radiative emissions from the sample were detected using a NIR InGaAs camera and a single element InGaAs detector. A long-pass filter LP-1000 nm was used to prevent the excitation laser beams from interfering with the InGaAs NIR camera and the detector. Diffuse radiative recombination-induced signals were collected and collimated with two off-axis paraboloidal mirrors focused on the detector. The detected signal was demodulated using a lock-in amplifier for the single-detector signal. In-house developed software was used for camera lock-in images referenced at the common frequency $f_1 = f_2$ in the case of homodyne LIC/PCR and at the (beat) frequency difference $\Delta f_{he} = |f_1 - f_2|$ for heterodyne LIC/PCR. The frequency difference Δf_{he} between the two beams used in the LIC and PCR measurements was 5 or 10 Hz for experiments. The samples were a *p*-type Si wafer with thick Si oxide (diameter: 150 mm, resistivity: 28–48 Ω cm, thickness 675 μm), an *n*-type Si wafer with thin silicon oxide (diameter: 100 mm, resistivity: 0.001–0.005 Ω cm, thickness 500 μm) and with passivation by an amorphous Si layer (diameter: 100 mm, resistivity: 1 Ω cm, thickness 290 μm), and an industrial multicrystalline *p*-type silicon solar cell (Si mc-SC) (156×156 mm^2, thickness 200 μm).

IV. RESULTS AND DISCUSSION

The experimental homodyne amplitude and phase frequency scans of the *p*-type and *n*-type Si wafers with the corresponding $|\Delta F_j(\omega)|$ and $\Delta\Phi(\omega)$ for two excitation intensities are presented in Fig. 4. Intensity I was approximately two times higher than intensity II. The $|\Delta F_j(\omega)|$ and $-\Delta\Phi(\omega)$ frequency CTG plots of both wafers exhibit one maximum. The frequency position of the maxima and the evaluated lifetimes are presented in Table III.

Small differences between lifetimes evaluated from amplitude and phase CTG are observed in all PCR measurements. Lifetimes decrease with increasing intensity due to the saturation of recombination mechanisms and enhanced carrier–carrier interactions.[18,19]

Data using the heterodyne (HePCR) modality applied to the same wafers are shown in Fig. 5. In this modality, the notch phenomenon[13,14,20] is often present when trap-state-related carrier density wave (CDW) dynamics is involved with nonlinear

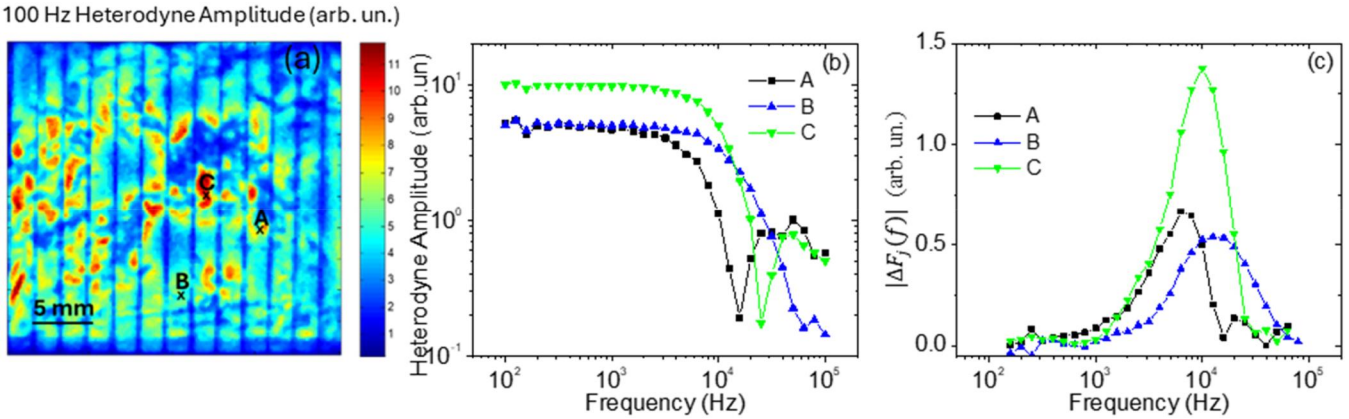


FIG. 7. (a) 100 Hz HeLIC amplitude image of Si mc-SC, (b) amplitude frequency responses of pixel A, B, and C, and (c) the corresponding CTG amplitudes $|\Delta F_j(\omega)|$.

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TABLE VI. CTG amplitude extrema frequencies and corresponding lifetimes and weighting factors for pixels A, B, and C (Fig. 7).

Pixel	k	$f_{ex}^{(A)}$ (Hz)	$\tau^{(A)}$ (s)	$f(\tau^{(A)})$
A	1.259	6.14×10^3	3.27×10^{-5}	1.92×10^5
		1.94×10^4	1.04×10^{-5}	-4.63×10^4
		5.89×10^4	3.41×10^{-6}	2.3×10^5
B	1.259	1.22×10^4	1.65×10^{-5}	…
C	1.259	9.08×10^3	2.21×10^{-5}	6.51×10^5
		3.26×10^4	6.16×10^{-6}	-1.21×10^5
		6.28×10^4	3.2×10^{-6}	7.78×10^4

electron–hole recombination interactions. A heterodyne frequency response notch is a resonant effect occurring when the modulation frequency matches the carrier trapping rate into, or emission rate from, energy traps in the bandgap. It should be mentioned that the characteristic "dip" (notch) shifts toward lower modulation frequencies as the optical intensity increases. As a result, the "notch" effect can reside at frequencies lower that the tested and displayed frequency range. The observed shift can be explained as follows:[21] At higher excitation intensities, a larger number of free holes are generated in the valence band. This decreases the probability of holes trapped in defect states to transition to available valence-band free hole states for recombination, thereby effectively

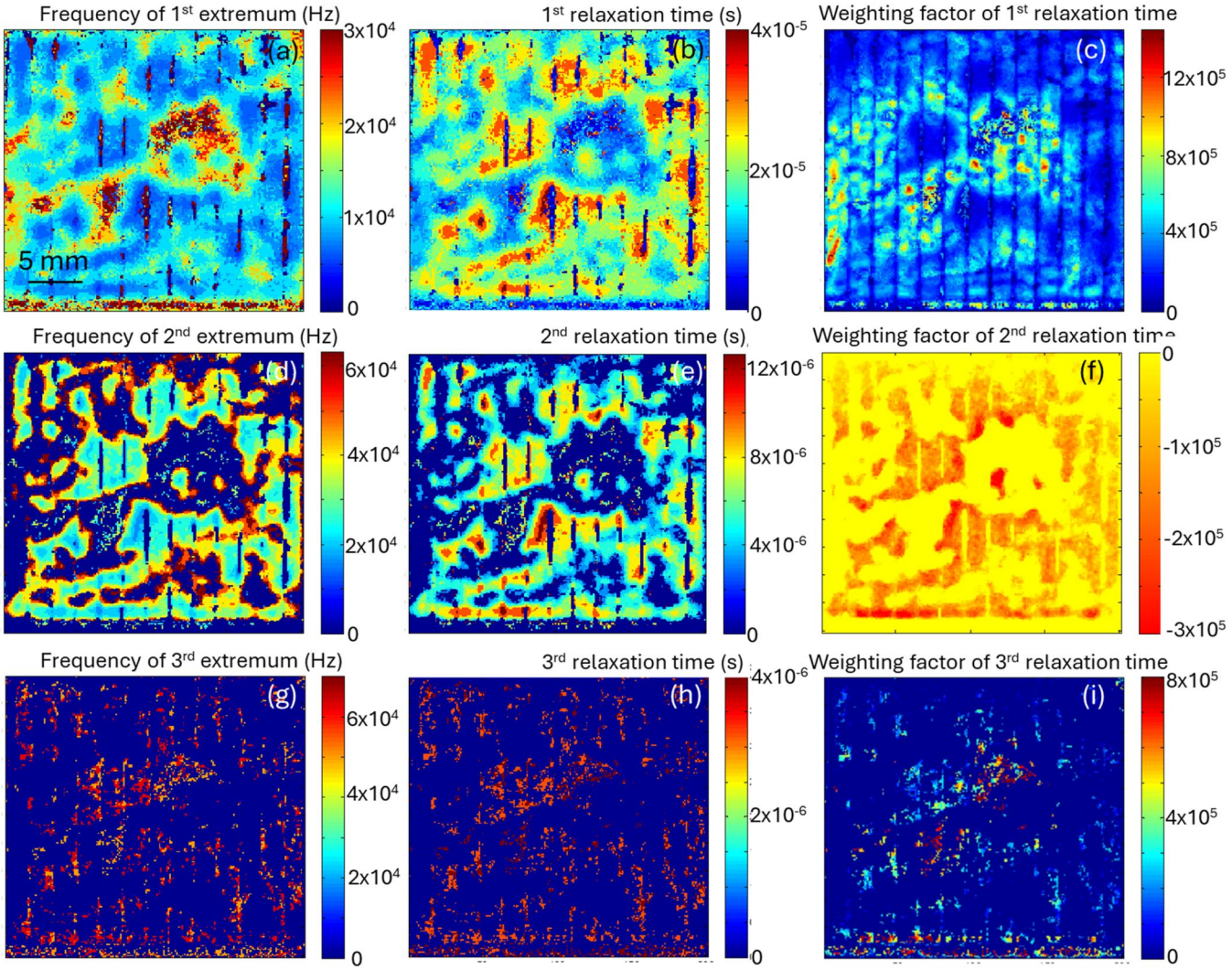


FIG. 8. CTG amplitude $|\Delta F_j(\omega)|$ frequency image of (a) the first extremum, (b) corresponding relaxation time, and (c) weighting factor; (d) frequency image $|\Delta F_j(\omega)|$ of the second extremum, (e) corresponding relaxation time, and (f) weighting factor; and (g) frequency image of the third extremum, (h) corresponding relaxation time, and (i) weighting factor.

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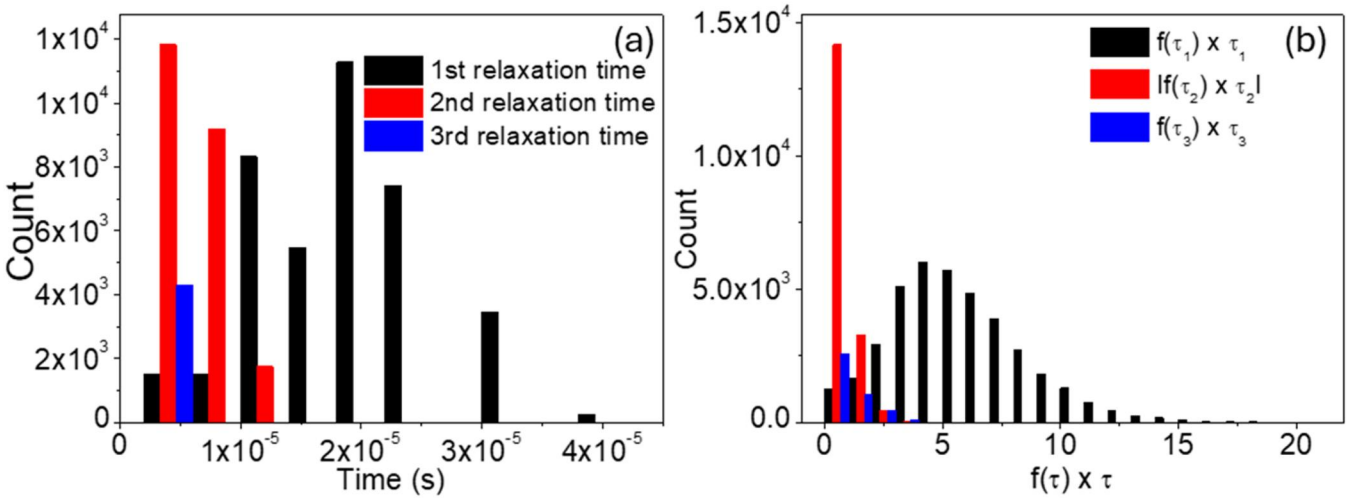


FIG. 9. Statistical distribution of (a) relaxation times and (b) $f(\tau) \times \tau$.

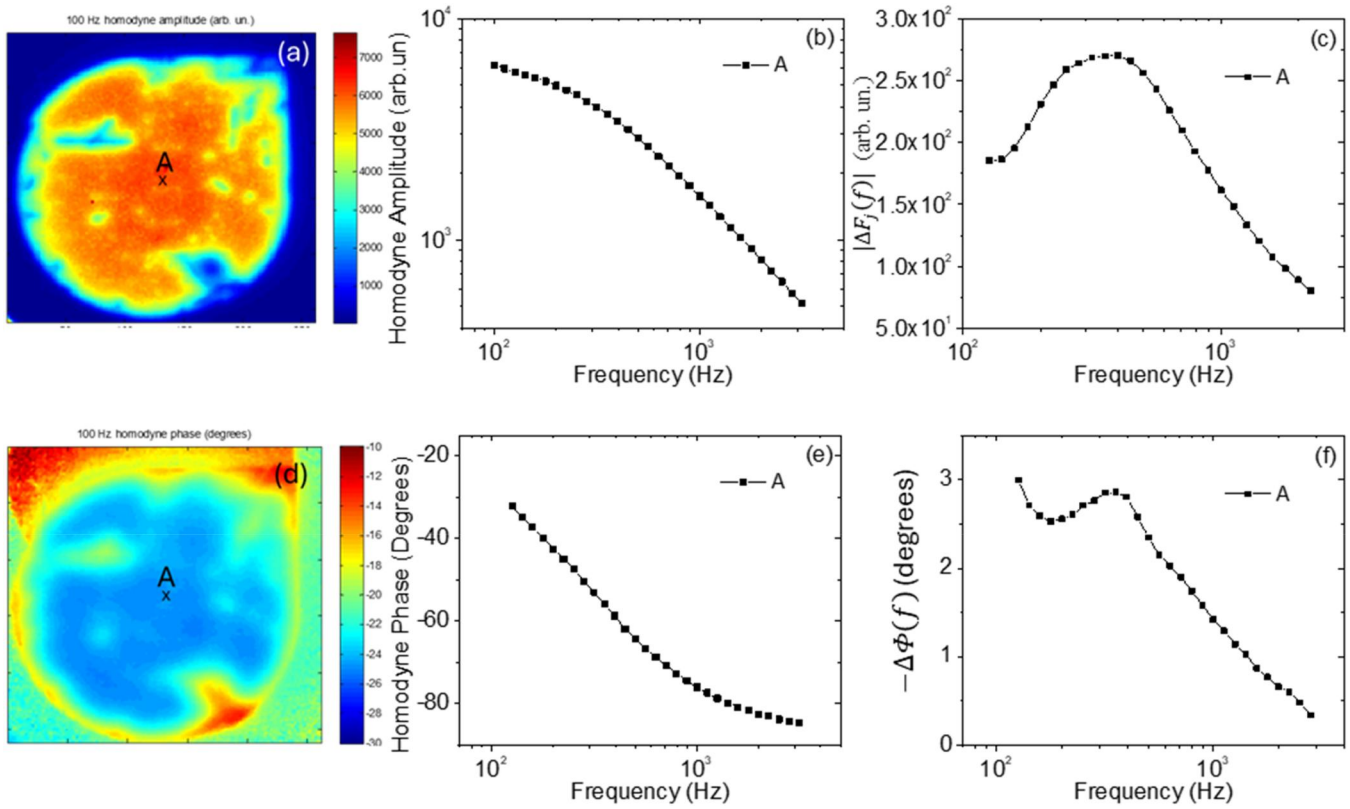


FIG. 10. (a) 100 Hz HoLIC amplitude and (d) phase images of a surface passivated *n*-type Si wafer; (b) frequency amplitude, and (e) frequency phase response of pixel A; (c) corresponding CTG $|\Delta F_j(\omega)|$ and (f) $-\Delta\Phi(\omega)$ scans.

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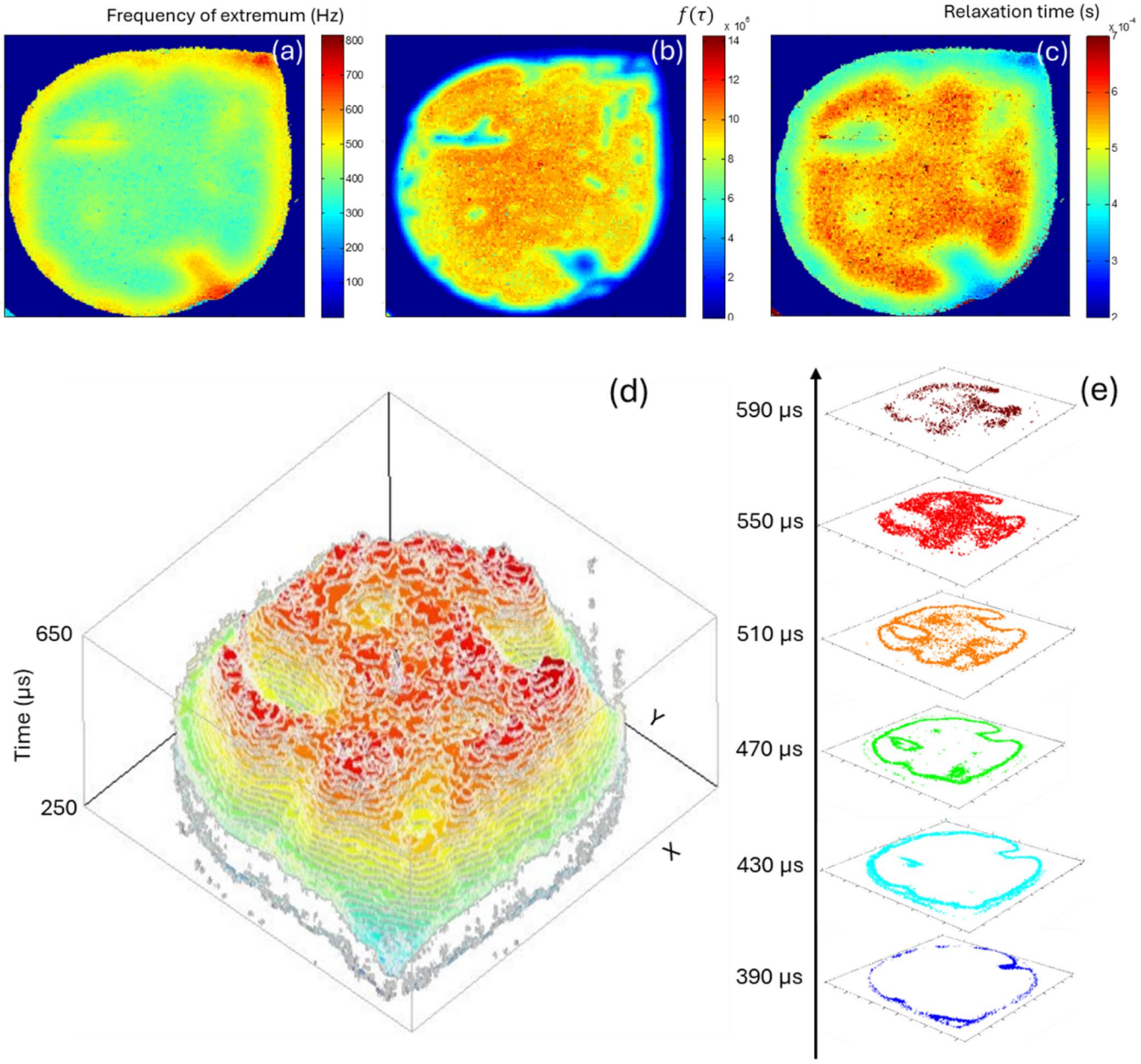


FIG. 11. Frequency distribution image of (a) the CTG amplitude extremum, (b) the corresponding weighting factor, (c) relaxation time, (d) three-dimensional chronocube with slice width 10 μs, and (e) slices with width 20 μs of the amorphous silicon passivated *n*-type Si wafer.

prolonging their trap residence time, reducing the capture/emission rate and thus shifting the notch to a lower frequency. This effect can be seen in the *p*-type Si wafer especially in the case of the low excitation intensity II.

In this case, the presence of an amplitude notch results in three or two CTG $|\Delta F_j(\omega)|$ extrema. Furthermore, $\Delta F_j(\omega)$ changes sign in the frequency range where the extremum is observed, which implies that at least one of the $f(\tau_j)$ factors in Eq. (14) is negative. This is a sign of carrier emission as opposed to capture, the latter process being consistent with the definition of positive $f(\tau_j)$ as a parameter associated with *carrier decay (loss) probability*. A negative $f(\tau_j)$ and the corresponding τ_j will, therefore, be

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associated with *carrier generation (gain) probability*. As a result, the extremum of $\Delta F(\omega^*_{(A)})$ [Figs. 5(e) and 5(f)] must take into account negative $f(\tau_j)$ values. This range corresponds to high emission rates from energy traps, and the associated relaxation time is probably affected by this effect. The frequency positions of the maxima and the evaluated lifetimes are presented in Table IV.

The lifetimes estimated from the heterodyne CTG amplitudes differ from those estimated from the homodyne CTG amplitude dependence. This is due to the different physical origins of the HePCR signals, which, unlike HoPCR signals, are sensitive to carrier kinetic rates from capture/emission processes into/out-of electronic traps.[13,21]

In contrast, HoPCR signals are predominantly sensitive to band-to-band radiative recombination rates.[1,22] HoPCR frequency scans of the *n*-type monocrystalline solar cell using different laser illuminated areas, and the associated CTG amplitude, $|\Delta F_j(\omega)|$, and phase, $-\Delta\Phi(\omega)$, frequency dependencies are presented in Fig. 6.

The appearance of an additional relaxation time at decreasing illuminated areas is observed very clearly in the CTG $|\Delta F_j(\omega)|$ and $-\Delta\Phi(\omega)$ frequency dependencies. The additional relaxation time is caused by the charge transport layer capacitance of the *p–n* junction,[22] which depends on the size of the laser illuminated area. The evaluated parameters are shown in Table V.

The results from single CTG amplitude and phase peak relaxation times are consistent with the (longer) lifetime corresponding to minority carrier recombination, while the shorter relaxation times from two CTG phase peaks are consistent with the RC time constant, which decreases in tandem with the decreasing charge transport layer capacitance with decreasing illuminated area.[22]

HeLIC imaging of the laser illuminated surface of the solar cell followed by CTG processing was further implemented at several modulation frequencies in order to yield enhanced images of all lifetimes across the sample.

Typical multicrystalline Si solar cells (Si mc-SC) display very inhomogeneous distributions of carrier lifetimes,[14] resulting in different areas exhibiting different HeLIC frequency responses. Figure 7 shows a 100 Hz HeLIC amplitude image, and HeLIC and CTG amplitude frequency scans for three pixels at different locations.

It is clear that two of the selected pixel responses exhibit the "notch" effect. The corresponding $|\Delta F_j(\omega)|$responses also exhibit different numbers of extrema with the calculated relaxation times and their weighting factors collected in Table VI.

In this case, the CTG frequency scan of pixel B (no notch) reveals only one lifetime, whereas those of pixels A and C (with notch) reveal three lifetimes. The former single lifetime is most certainly associated with minority radiative recombination, which usually dominates radiative emissions, whereas the latter three lifetimes have been found to be due to radiative recombination τ_n and τ_p on both sides of the p–n junction, plus the charge transport layer RC time constant, τ_{pRC}.[22]

Using the foregoing procedure, CTG-enhanced multi-lifetime images of the solar cell and the corresponding weighting factors were constructed from HeLIC imaging data (Fig. 8).

As shown in Table VI, a consistent trend of negative weighting factors was found associated with the middle (second longer) relaxation times when a frequency notch was observed. These images show wide differences in, and numbers of, relaxation lifetimes at different areas of the Si mc-SC.

The statistical distribution of relaxation times is shown in Fig. 9. It should be mentioned that the statistical distribution of $f(\tau)\times\tau$ is more valuable than that of τ alone because it compares the *relative impact* of each lifetime to other lifetimes, by weighing

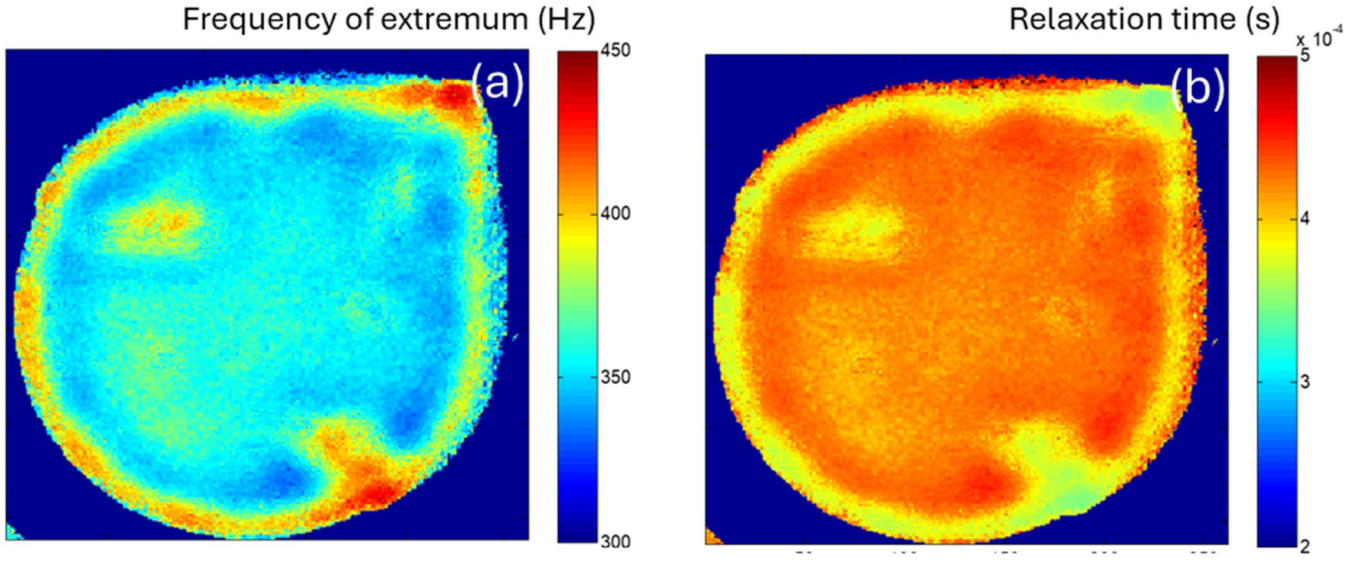


FIG. 12. Frequency distribution image of (a) the CTG $-\Delta\Phi(\omega)$ peak; and (b) the corresponding relaxation times of the amorphous silicon passivated *n*-type Si wafer.

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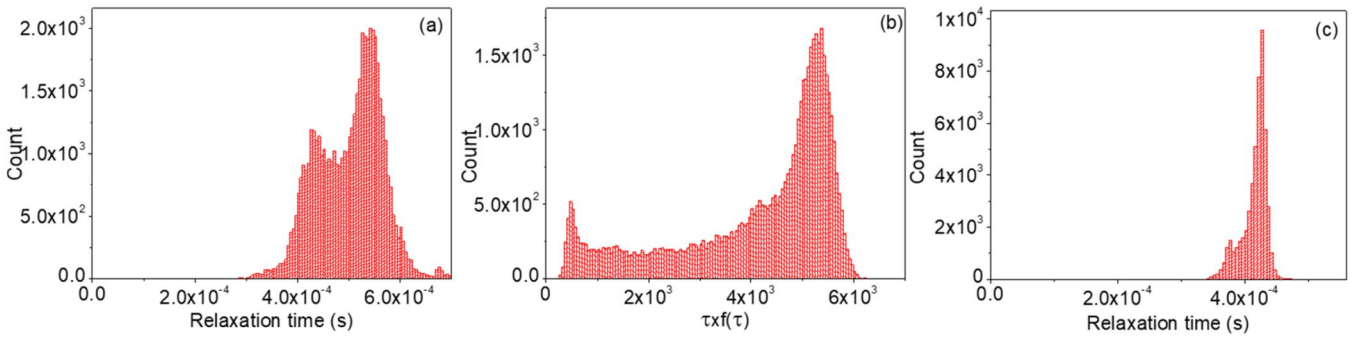


FIG. 13. Statistic distribution of lifetimes evaluated from (a) the CTG amplitude, (b) the corresponding $f(\tau)\tau$, and (c) phase $-\Delta\Phi(\omega)$ peak of the amorphous silicon passivated *n*-type Si wafer.

the value of each lifetime with the relative probability of its occurrence.

Figure 9(b) shows that CTG is an important analysis tool for semiconductor optoelectronic quality control, as it reveals for the first time to the best of our knowledge the efficient carrier loss mechanisms with short lifetimes (τ_3, blue histogram; and τ_2, red histogram), which control fast decay rates with high probability (red histogram bars), operating hidden below much longer, more readily measurable lifetimes (τ_1, black histogram bars).

The homodyne (HoLIC) modality was also studied as a means of generating CTG images. The 100 Hz HoLIC amplitude image of another *n*-type Si wafer passivated with amorphous silicon, and its frequency amplitude and phase dependencies and those of the corresponding CTG $|\Delta F_j(\omega)|$ and $-\Delta\Phi(\omega)$ of a camera pixel are shown in Fig. 10. A much more uniform degree of homogeneity across the wafer is shown in Fig. 10(a) with sharp variations mostly along the near-rim regions.

The CTG amplitude response, $|\Delta F_j(\omega)|$, as well as phase response, $-\Delta\Phi(\omega)$, at the centerpoint A of the wafer exhibits only one peak. The frequency distribution of the amplitude CTG maxima and the derived weighting factor and relaxation times are presented in Fig. 11. The CTG lifetime image, Fig. 11(c), exhibits a great degree of long lifetime uniformity due to radiative recombination along all regions away from the rim. Near the rim, shorter relaxation times emerge along with a few regions radially closer to the center. These are single CTG lifetime regions and can be interpreted as having short radiative recombination times with likely higher non-radiative-decay rates. The observed weighting factor inhomogeneity [Fig. 11(b)] may be caused by different recombination mechanisms in the investigated areas.

The frequency distribution of the CTG $-\Delta\Phi(\omega)$ maximum and the derived relaxation time are presented in Fig. 12. The derived relaxation time image is very similar to that derived from the CTG amplitude, as expected. The statistical distributions of lifetimes over the entire wafer area are presented in Fig. 13.

The lifetime distribution has much smaller linewidth than in the case of Si mc-SC [Fig. 9(a)]. It is also seen that the phase lifetime distribution is much sharper than in the case of the amplitude. This is expected because the PCR phase [Fig. 13(c)] is much more sensitive to decay lifetimes than the PCR amplitude [Fig. 13(a)]. The statistical distribution of $f(\tau)\times\tau$ [Fig. 13(b)] enhances the short (hidden) lifetime range below 3×10^{-4} s, showing a high probability of fast carrier loss events, which will impact device performance when fabricated on this passivated *n*-type Si wafer.

V. CONCLUSIONS

The concept of frequency-domain chronotomography (CTG) was introduced, and semiconductor multi-lifetime PCR and LIC imaging applications were presented. CTG is a scanning adaptive frequency gate, a resonant process that significantly enhances the PCR signal response and LIC image contrast and resolution to the presence of one or more free-carrier lifetime limiting mechanisms when the free-carrier decay rates match/lie within the frequency gatewidth between two successive modulation frequencies (a frequency gate). The new CTG modality allows the measurement of multiple relaxation times (decay lifetimes) and their probability of occurrence directly from experimental frequency PCR scans, thereby avoiding theoretical fitting procedures. It was shown that CTG features high sensitivity to multi-lifetime spectra, including usually hidden spatial distributions and their occurrence probabilities in thus characterized Si wafers and solar cells.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

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Author Contributions

Andreas Mandelis: Conceptualization (equal); Formal analysis (equal); Funding acquisition (equal); Investigation (equal); Methodology (equal); Project administration (equal); Supervision (equal); Validation (equal); Writing – original draft (equal); Writing – review & editing (equal). **Alexander Melnikov:** Data curation (equal); Formal analysis (equal); Investigation (equal); Methodology (equal); Validation (equal); Writing – original draft (equal). **Angelina (Xinyi) Zhang:** Data curation (equal); Formal analysis (equal); Methodology (equal). **Qiming Sun:** Formal analysis (equal); Methodology (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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